GlycoTwin

Personalized Post-Meal Glucose Digital Twin

Digital Twin Challenge 2026

A stateful, uncertainty-aware Digital Twin that forecasts post-meal glucose risk, learns from each person's own eligible outcomes, and reports its evidence honestly — rather than issuing a single prediction and forgetting it.

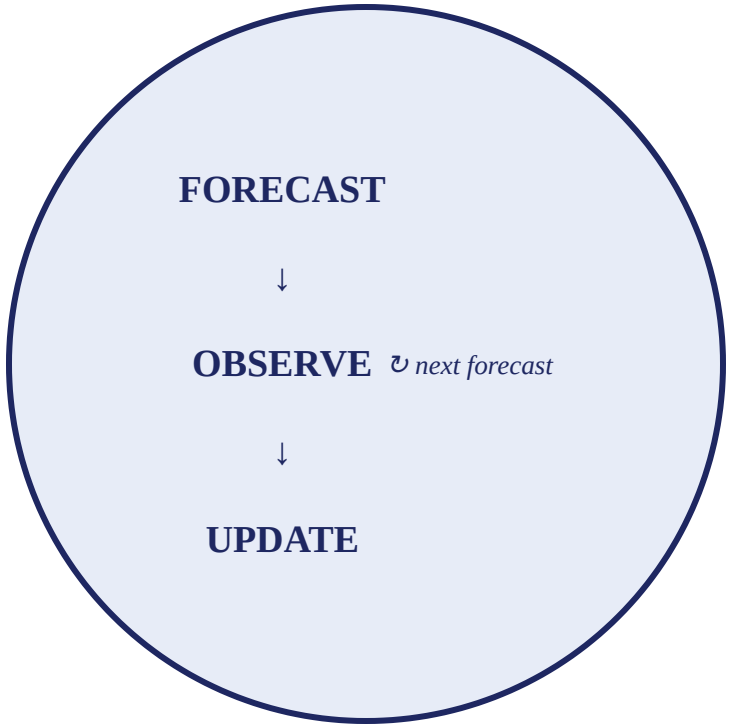
FORECAST
with uncertainty

→

OBSERVE
eligible outcome

→

UPDATE
new twin version



Team: NeuralForge — Medha Choudhari (solo)

Institution: Bapuji Institute of Engineering and Technology (BIET), Davangere, Karnataka

Incubator: *[Pending confirmation]*

Problem Statement

THE PROBLEM

Post-meal glucose responses vary from person to person — age, body composition, recent activity and individual metabolism all matter. A single population-average forecast misses this variation entirely.

CURRENT LIMITATION

A one-time prediction does not automatically adapt when new, person-specific observations become available. Most tools also present an estimate without reporting how confident they are.

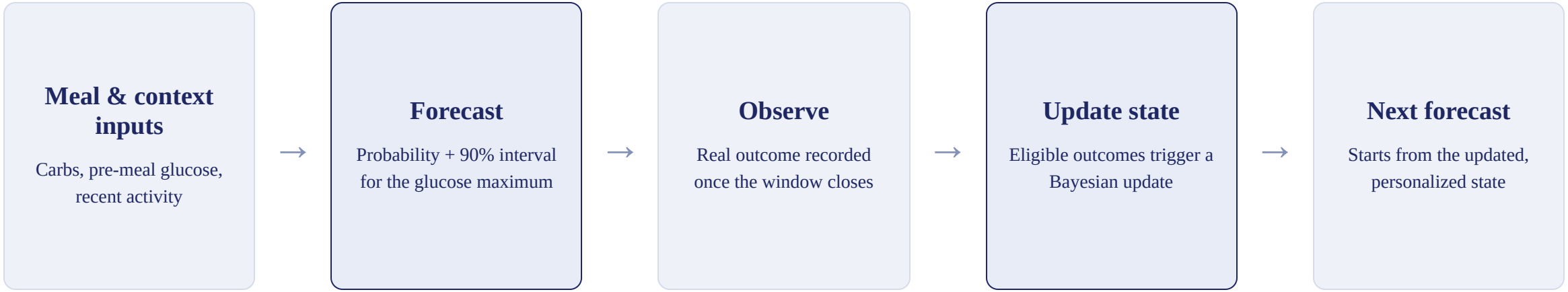
CENTRAL RESEARCH QUESTION

Can a stateful, personalized Digital Twin connect forecasts with eligible observations, and use them to update itself for the next forecast?

- Personalization and honest communication of uncertainty are both important research considerations — not optional extras for GlycoTwin's design.
- Prior single-shot predictors treat every forecast as independent; GlycoTwin treats each forecast as one step in a continuing, evidence-linked loop.

Proposed Solution

GlycoTwin keeps a persistent, versioned Digital Twin for each person and runs it through one continuous workflow:



The twin's state carries forward — the next forecast starts from the updated parameters, not from scratch. Each stage above is logged, so any forecast can be traced back to the twin version that produced it.

The prediction target is the **maximum** glucose reached within the 120-minute post-meal window — not necessarily a complete glucose-over-time curve. GlycoTwin forecasts a single risk value with uncertainty, not a simulated trajectory. This keeps the modelling problem well-posed even when continuous glucose readings are noisy or sparse.

Innovative Features

★

Forecast → Observe → Update — the core loop

Connects every prediction with its eligible observed outcome and the state update that follows — a forecast is never a dead end. This is the project's central contribution.

1

Uncertainty-aware forecasting

Reports a 90% interval and a threshold probability — never presents an estimate as a guarantee.

2

Bayesian-style personalization

Model B updates a personal carbohydrate-response parameter from each person's own eligible observations.

3

Persistent Digital Twin state

Retains model state and full version history, carried into every subsequent forecast.

4

Explainability & model comparison

Lets researchers inspect outputs and compare Models B and C side by side, honestly.

The project's contribution is the integrated, working loop — not any single technique in isolation. Each supporting feature above strengthens the same forecast→observe→update cycle shown at the top of this slide.

Dataset and Research Methodology

CGMacros dataset — **45 participant files audited**. Figures below trace one pipeline, stage by stage:

1,706
logged meal rows

→

1,697
valid 120-min windows (extraction-valid)

→

1,262
core-eligible events (leakage-safe modelling set)

→

963 / 34
activity-eligible events / participants (Model B vs C)

- Meal event anchored at the logged meal row (t0); outcome window is (t0, t0+120 min].
- Features use only information available before the meal; activity measured strictly before t0.
- Leakage safeguards: participant-level folds, sequential (online) evaluation.
- Not every model used every event — 963/34 is the activity-eligible subset used specifically for Model B vs C.

*No participant-level health data or the dataset itself is included in this repository — aggregate, de-identified figures only. All counts above are reproducible from the audited pipeline in **data/audit**.*

System Architecture and Technical Stack

Frontend

React 19 · TypeScript · Vite
7 pages, calls /api endpoints

→

Backend API

FastAPI, bound to 127.0.0.1 only
Session cookie + ownership check on every twin route · scrypt password hashing

→

Database

SQLite — 11 tables
users, twins, forecasts, observations, reconciliations…

LIVE APPLICATION — research engine (shared)

Bayesian Models B & C — closed-form update, forecast + uncertainty. The live twin and offline research experiments call the exact same functions, so a result validated offline reflects the engine actually serving users.

OFFLINE — CGMacros research pipeline

Runs on the researcher's own machine — never reached by the website. Only aggregate priors and evaluation reports cross over into the live application, never raw participant data.

Technical stack, grouped by layer:

- pandas, NumPy, SciPy, scikit-learn, XGBoost (Model A) — src/glycotwin/models/
- FastAPI, Pydantic v2, SQLAlchemy 2, Uvicorn, SQLite — src/glycotwin/backend/
- pytest, httpx for testing (815 Python tests recorded)

Separating the live and offline paths keeps the deployed application simple and auditable, while still letting the same Bayesian engine be stress-tested against a full retrospective dataset.

ML Models and Personalization

Model	Role	Formula	Personalizes / updates?
A	Population baseline (XGBoost)	tree ensemble on pooled data	No — trained once, no personal history
B	Personalized carb-sensitivity model	rise = β · carbs + noise	Yes — β updates after each eligible meal
C	Model B + activity conditioning	rise = (β + γ·activity) · carbs + noise	Yes — same update, plus γ term

- In plain English: β ("beta") is how much this person's glucose typically rises per gram of carbohydrate; γ ("gamma") adjusts that for recent activity.
- Both parameters start from a population prior and are refined per person as eligible meals accumulate.
- After an eligible outcome, a closed-form Bayesian update moves β (and γ, for Model C) toward what that meal suggests and narrows its uncertainty.
- Model A serves as the fixed, non-personalized baseline that B and C are compared against.

This is a parameter update, not retraining a neural network — no model is retrained from scratch after each meal.

Research Results and Evaluation

Model B vs Model C — 963 meals, 34 participants, Libre channel, out-of-fold + sequential evaluation:

Metric	Model B	Model C	C − B	95% interval (illustrative)
ROC-AUC	0.7943	0.8059	+0.0116	−0.0084  +0.0479
Brier score	0.1391	0.1379	−0.0012	−0.0064  +0.0030
Rise MAE (mg/dL)	29.40	29.51	+0.11	−0.52  +0.68

8 of 9 paired intervals include zero (bars crossing the navy zero-line above) → the comparison is INCONCLUSIVE

Model B does not demonstrably outperform every baseline; Model C's activity term has not been shown to help. No universal superiority is claimed for either model — the honest reading of this evidence is "not yet distinguishable," not "Model C wins." Reporting this plainly, rather than rounding it up to a win, is itself part of the project's evidence discipline.

Three different kinds of evidence: this retrospective CGMacros result is research evidence — not the same as the live application working (next slide), and neither is the same as clinical validation, which has not been done. All three are necessary; none substitutes for another.

Demonstration, Limitations and Safety

VERSION 0 (before)

7.7%

P(max ≥ 180 mg/dL)

→

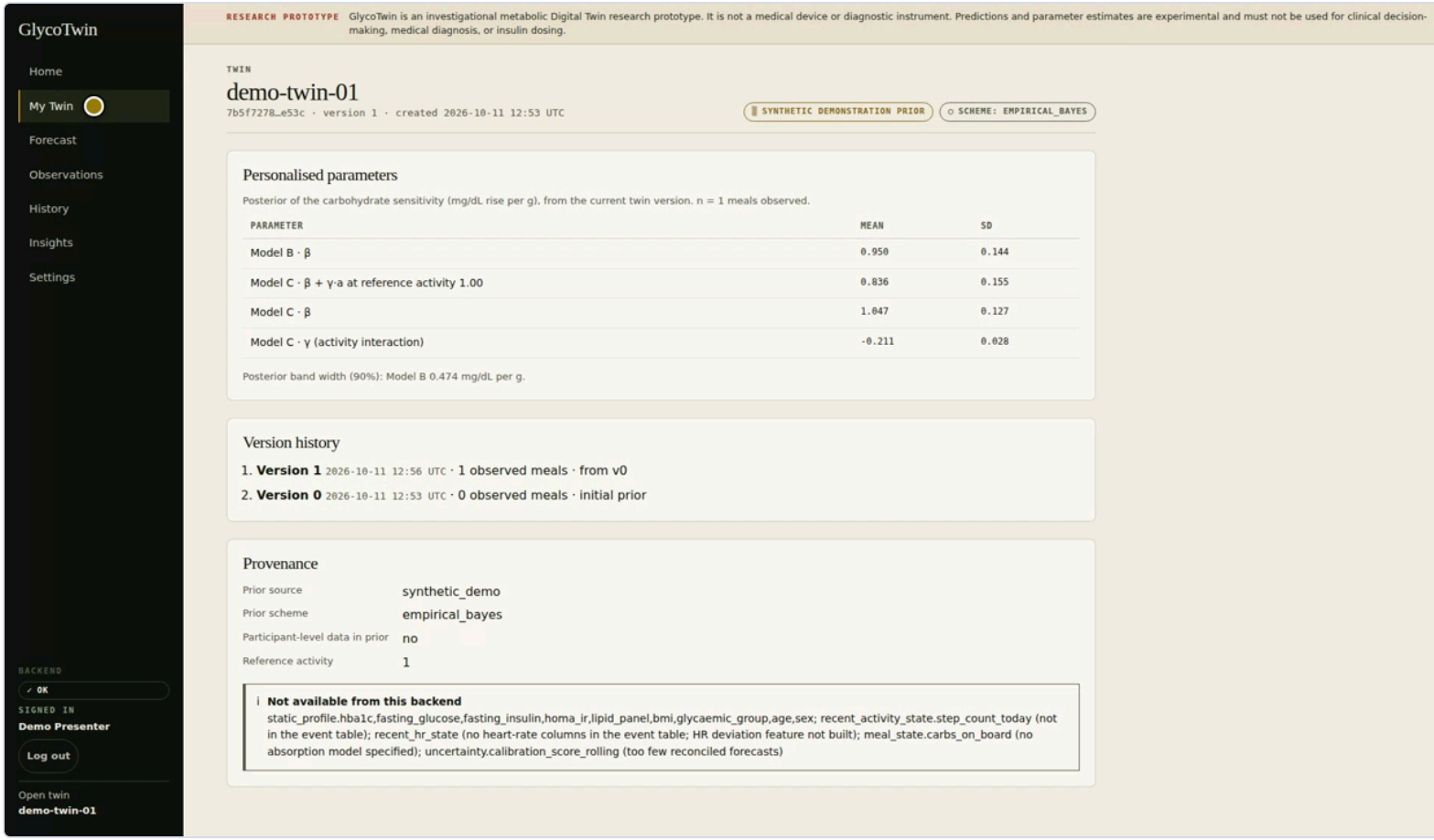
VERSION 1 (after)

21.0%

P(max ≥ 180 mg/dL)

- Observe: eligible outcome recorded after the 120-min window closes
- Update: Model B β moves from 0.747 → 0.950 (twin version 0 → 1)
- Next forecast, same inputs: probability rises to 21.0%

Same inputs, different twin state, different forecast.



Real application screenshot — My Twin page after one eligible update (version 0 → 1)

Limitations and safety

- Investigational research prototype — not a clinically validated diagnostic or treatment system.
- Not intended for insulin dosing or any clinical decision.
- Further evaluation and independent validation are required before any real-world use.

Conclusion and Future Work

PROBLEM

One-off glucose predictions cannot adapt to the individual, and most tools do not report how confident they are.

KEY CONTRIBUTION

A persistent, versioned Digital Twin linking forecasting, uncertainty and sequential Bayesian personalization in one working loop.

RESEARCH EVIDENCE

Retrospective CGMacros evaluation — inconclusive for activity-conditioning, but the pipeline and leakage safeguards are validated.

NEXT STEPS

Run the full prespecified key experiment on real data with baseline controls; test whether an activity effect depends on activity level.

Repository: github.com/medhachoudhari/GlycoTwin

Team / Institution: NeuralForge — Medha Choudhari, BIET Davangere

GlycoTwin connects forecasting, observation, and personalization into one honest, auditable research workflow — reporting what the evidence shows, not more.

Investigational research prototype — not a medical device, not clinically validated.